



Sequence **partitioning**

Andrea Guarracino
Assistant Professor



@AndresGuarahino



EmboPanG26

Torre del Greco, NA, Italy
2026/06/11

**GUARRACINO
LAB**

<https://guarracinolab.github.io>

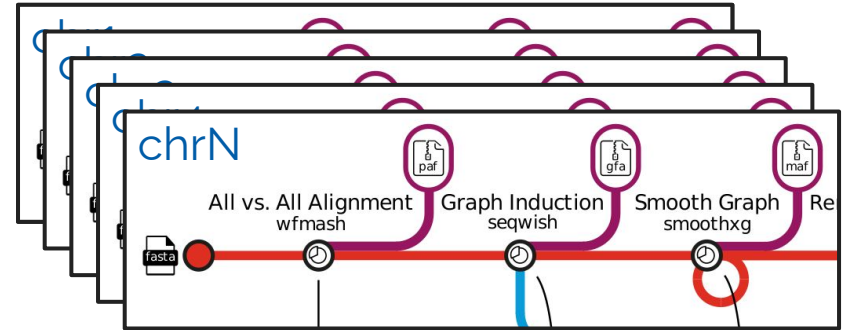


Outline

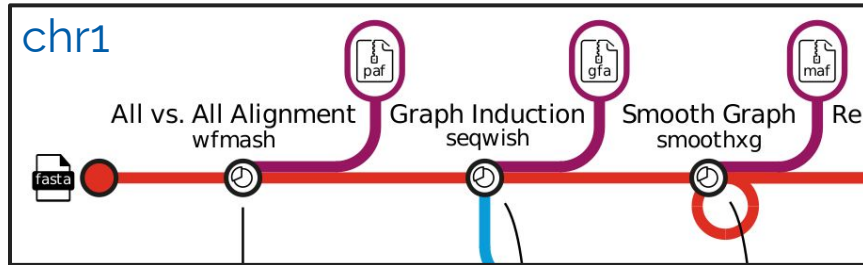
- Reference-guided partitioning
- Reference-free partitioning

Outline

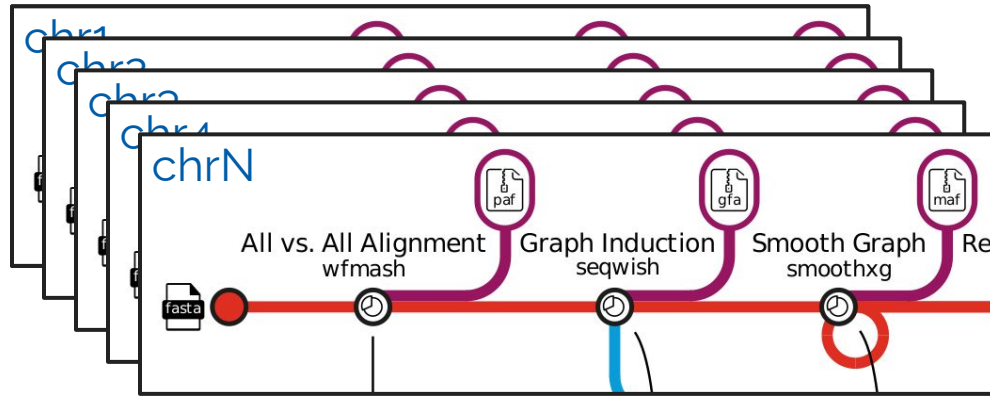
- Reference-guided partitioning
- Reference-free partitioning



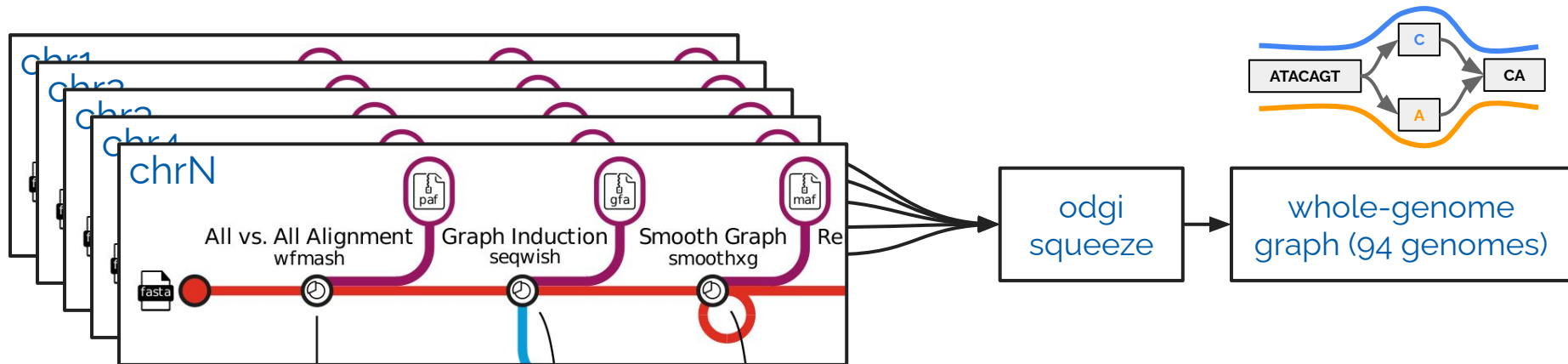
Reference-guided sequence partitioning



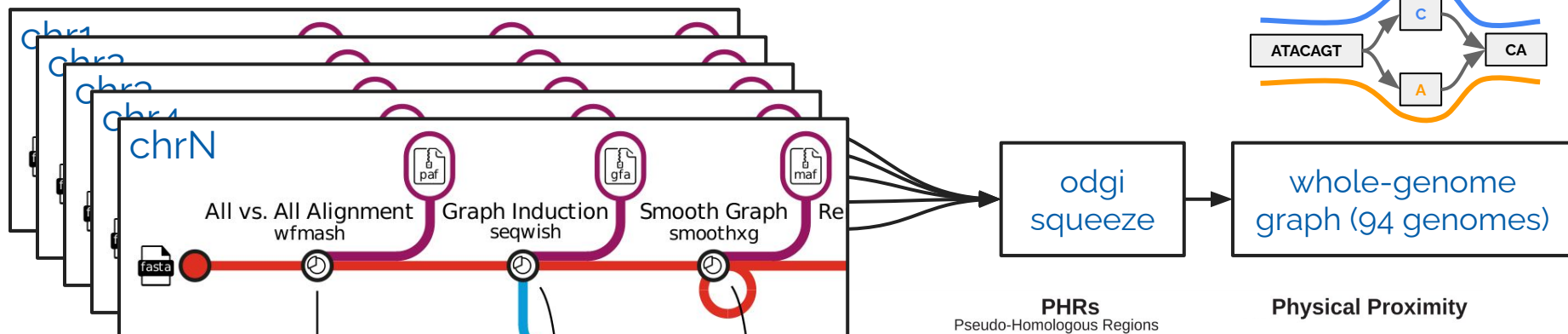
Reference-guided sequence partitioning



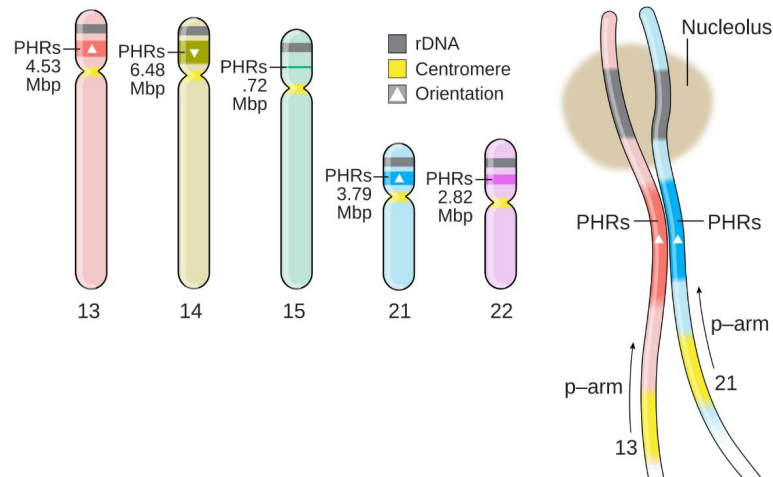
Reference-guided sequence partitioning



Reference-guided sequence partitioning

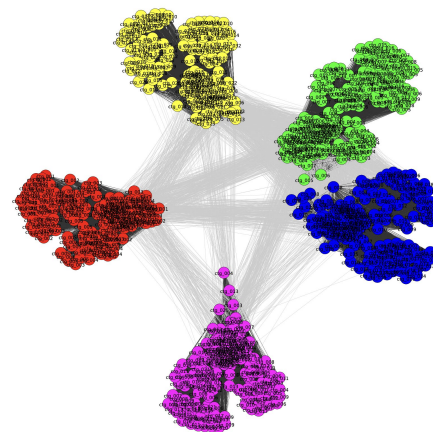


This approach has limitations
(see [Guarracino et al., 2023, Nature](#))

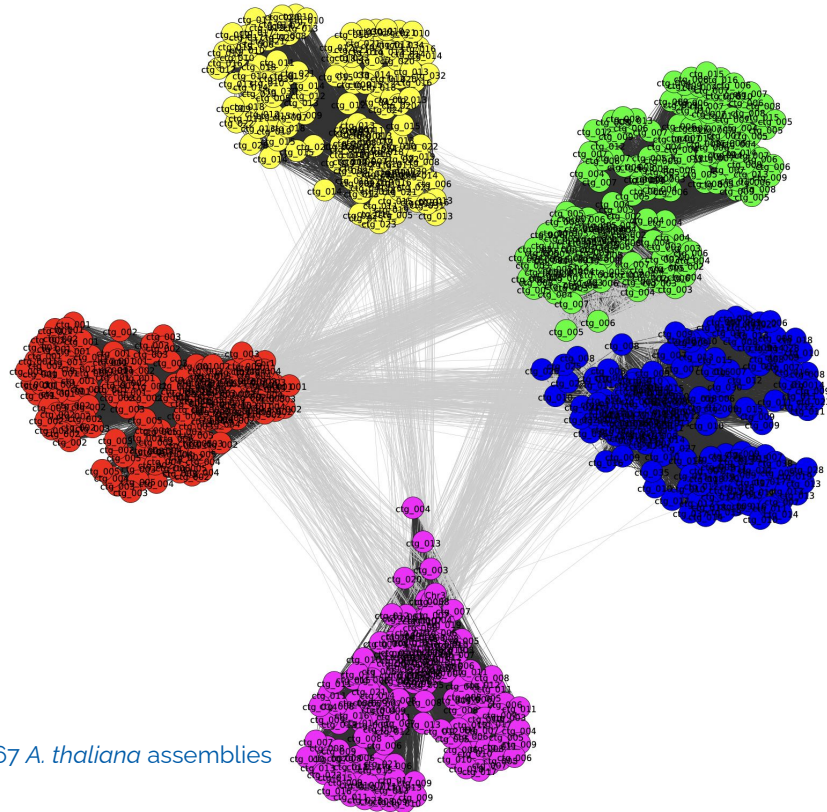


Outline

- Reference-guided partitioning
- Reference-free partitioning

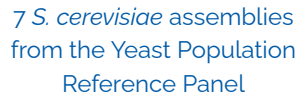


Reference-free sequence partitioning

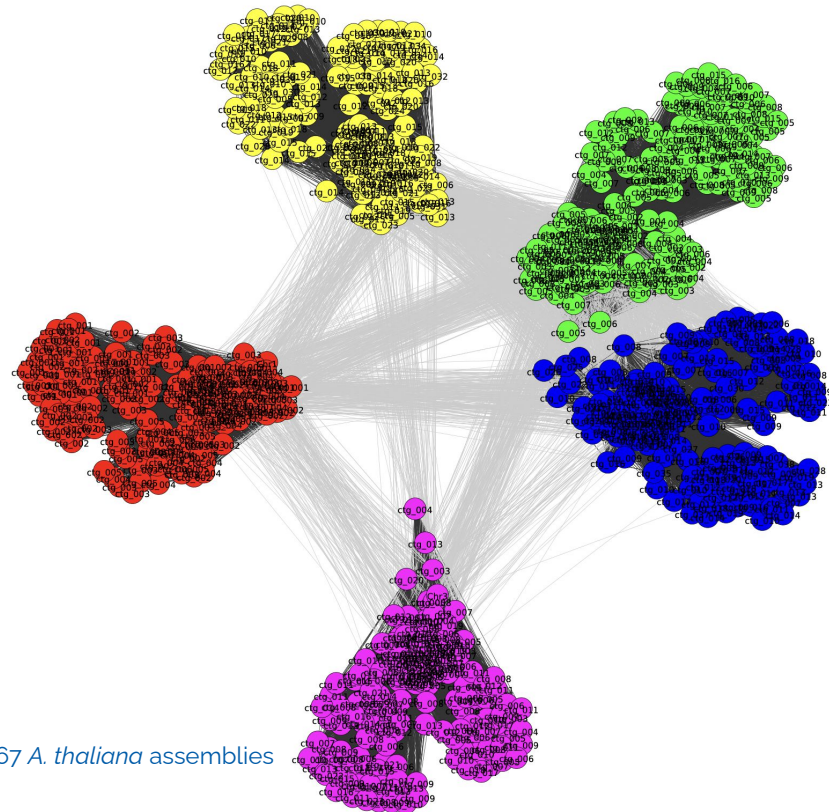


67 *A. thaliana* assemblies

Reference-free sequence partitioning

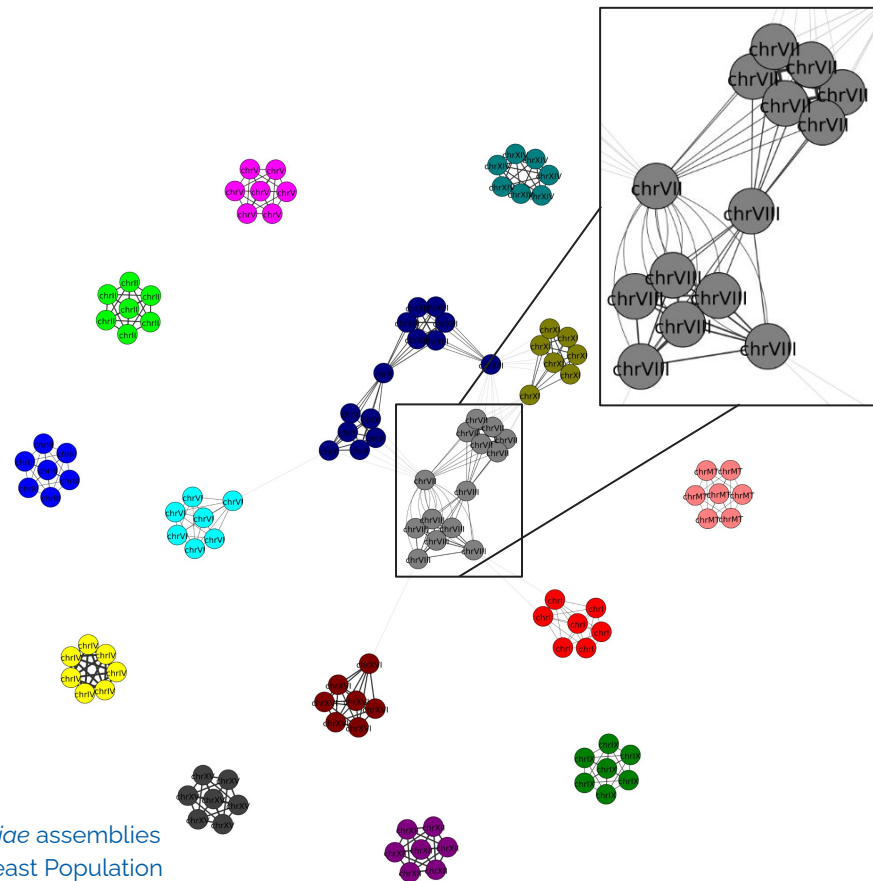


Reference-free sequence partitioning

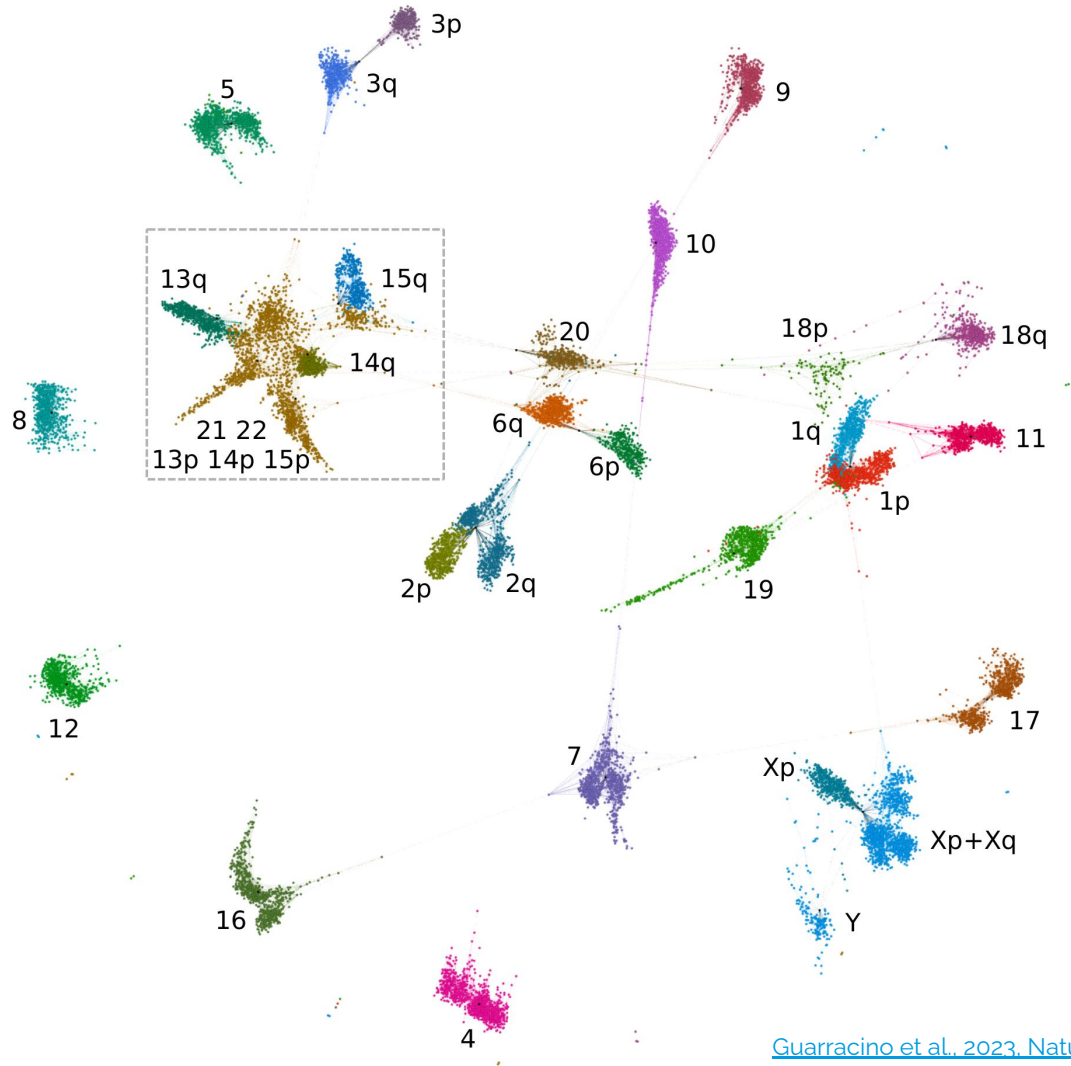


67 *A. thaliana* assemblies

7 *S. cerevisiae* assemblies
from the Yeast Population
Reference Panel



Chromosome communities in the HPRC



A simplified mapping
graph of contigs >1Mbps.

Questions?